

NWSL Pythagorean Points Prediction Model

An R implementation of the Pythagorean expectation model for the National Women's Soccer League (NWSL), adapted from Bill James' original baseball formula. The model estimates how many points a team *should* have based on their goals scored and conceded, identifies over- and underperforming teams, and projects final season standings from a mid-season snapshot.

Background

The Pythagorean expectation estimates a team's expected points percentage as:

'''

$$\text{ExpPointsPct} = \text{GF}^x / (\text{GF}^x + \text{GA}^x)$$

'''

Where `GF` is goals for, `GA` is goals against, and `x` is an exponent fitted to the data. In baseball the exponent is typically around 1.83. For soccer, this model fits `x` directly from historical NWSL season data — yielding a value of around **1.43**.

Points percentage is used instead of win percentage to correctly handle draws (which soccer has and baseball does not).

Scripts

`nwsl\_pythagorean\_simple.R`

The simplest version. Takes a CSV with match-level data (or a team-totals CSV), computes total GF and GA per team, and adds a single `PythExpPct` column using the exponents from Clive Beggs, based on EPL matches. Beggs' formula is:

'''

$$\text{ExpPointsPct} = a [\text{GF}^b / (\text{GF}^c + \text{GA}^d)] \times m$$

Where: GF is Goals For, GA is Goals Against, m is the number of matches, and a , b , c , and d are empirical coefficients. Using Beggs' coefficients: $a = 2.78$, $b = 1.24$, $c = 1.24$, $d = 1.25$.

'''

nws1_pythagorean_full.R

The full model. Everything in the single-season version, plus:

- Home/away split goal rates
- Match-by-match remaining schedule projection (inferred from the round-robin structure)
- Full SoS adjustment applied consistently to both expected points and projections

Inputs: same as `nws1_pythagorean_single_season.R`

****Input:**** ``Date``, ``Home Team``, ``Away Team``, ``Home Goals``, ``Away Goals``

How it works

1. Exponent fitting

Rather than borrowing a literature value, x is fitted from historical NWSL season-totals data using least squares — minimizing the squared error between predicted and actual points percentage across all team-seasons. This gives an NWSL-specific estimate rather than one derived from men's leagues.

2. Strength of Schedule (SoS)

An SRS-style (Simple Rating System) iterative adjustment is used:

- Each team starts with their raw goal differential per game
- Each iteration computes the average rating of opponents faced and adds it to the team's raw GD
- Ratings are mean-centred each iteration to prevent drift
- After 50 iterations the ratings converge to a stable value in units of *goals per game above/below a league-average team*

Positive SoS = tougher schedule than average. Negative SoS = easier schedule.

3. Two-factor SoS adjustment

SoS is applied in two ways, both controlled by `sos\_weight` (default: 0.25):

****Goal quality adjustment:**** teams that compiled their numbers against strong opponents get a boost to their adjusted GF and a reduction to their adjusted GA — and vice versa for teams that faced weak opponents:

```r

Adj\_GF = GF\_pg + SoS\_Rating \* sos\_weight

Adj\_GA = GA\_pg - SoS\_Rating \* sos\_weight

```

****Points expectation adjustment:**** more points are expected when facing weak opponents, fewer against strong ones. Applied as a direct shift to expected points after the Pythagorean calculation.

4. Remaining schedule projection (full model only)

The NWSL uses a full home-and-away round robin, so the complete fixture list is deterministic from the team list alone. Played matches are subtracted to recover remaining fixtures. For each remaining match, expected goals are projected using:

- The home team's home GF/GA rate vs. the away team's away GF/GA rate
- Both adjusted by each team's SoS rating $\times$ `sos\_weight`

A draw probability is estimated from the closeness of projected goal totals and used to convert win probabilities into expected points. Projected remaining points are added to actual points already earned.

Configuration

All tuneable parameters are in Section 0 at the top of each script:

Parameter	Default	Description
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```
|---|---|---|
| `matches_file` | `"NWSL2026 Matches.csv"` | Path to current season
match data |
| `historical_files` | 2024 + 2025 CSVs | Paths to historical season-
totals CSVs used for exponent fitting |
| `total_games` | `30` | Full season length (NWSL expanded to 30
games in 2026) |
| `fallback_x` | `1.7` | Literature exponent (Hamilton 2011) used as
sanity check |
| `sos_iterations` | `50` | Number of SRS iterations (50 is sufficient for
convergence) |
| `sos_weight` | `0.25` | Fraction of SoS rating applied in adjustments (0
= no SoS, 1 = full) |
```

```
### NWSL season lengths by year
```

```
```r
season_length <- c(
 `2021` = 24,
 `2022` = 22,
 `2023` = 22,
 `2024` = 26,
 `2025` = 26,
 `2026` = 30
)
```
```

```
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```

```
## Data format
```

```
### Match-level data (^NWSL2026 Matches.csv`)
```

```
```
```

```
Date,Home Team,Away Team,Home Goals,Away Goals
2026-03-15,Washington Spirit,Portland Thorns,2,1
2026-03-15,Utah Royals,Gotham FC,0,0
```

```
...
```

```
```
```

```
### Historical season totals (nwsf_historical_2024.csv, etc.)  
...
```

```
Season,Team,Matches,Points,GF,GA  
2024,Washington Spirit,26,52,38,21  
2024,Portland Thorns,26,44,33,24
```

```
...  
...
```

```
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```

Requirements

- R (≥ 4.1)
- `tidyverse`
- `ggrepel`

Install with:

```
```r  
install.packages(c("tidyverse", "ggrepel"))
```
```

```
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```

Output

- **Console:** fitted exponent, R^2 , team-level expected vs actual points table, projected final standings
- **Plot:** `actual\_vs\_expected\_points.png` — scatter of actual vs SoS-adjusted expected points with team labels; points above the diagonal are overperforming, below are underperforming

```
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```

References

- James, B. (1980). *Baseball Abstract*. — original Pythagorean expectation

- Hamilton, H. (2011). An Extension of the Pythagorean Expectation for Association Football. *Journal of Quantitative Analysis in Sports*, 7(2).
- Morey, D. — Simple Rating System methodology (via Pro Football Reference)